VET-FACING COMPETITIVE COMPARISON

equicenta® CTM vs. Arthramid® Vet

Two very different products, often considered for the same cases. This sheet lays out how they compare on mechanism, indications, evidence base, and the practical decisions vets and clients care about.

KEY REFERENCE

Bertone AL, et al. Cryopreserved umbilical allograft (equicenta® CTM) improves clinical outcomes compared with clinician-selected standard care in performance horses. Frontiers in Veterinary Science (2026).
DOI: 10.3389/fvets.2026.1833933

	equicenta® CTM Cryopreserved equine umbilical cord tissue matrix (biologic allograft)	**Arthramid® Vet** 2.5% injectable polyacrylamide hydrogel (2.5% iPAAG, synthetic device)
Category	**equicenta® CTM**	**Arthramid® Vet**
Product class What it is	Biologic allograft. Cryopreserved equine umbilical cord tissue matrix containing native extracellular matrix proteins, growth factors, and extracellular vesicles.	Synthetic medical device. 2.5% polyacrylamide hydrogel — a cross-linked polymer scaffold in 97.5% sterile water; no biologic components. (Arthramid product literature)
Mechanism of action Primary MOA	Biologic: supports tissue repair and modulates inflammation. Multi-modal payload of native ECM proteins, growth factors, and small “messenger” particles released as the matrix biodegrades.	Biomechanical: integrates into the synovial membrane over ~2–4 weeks, creating a scaffold intended to reduce joint-capsule stiffness. Inert, non-pharmacologic. (Tnibar 2022; Arthramid label)
Site of action Where it works in the joint	Acts across joint tissues (synovium, cartilage matrix, peri-articular soft tissue) via biologic signaling; also used peri-tendinously and peri-ligamentously.	Acts on the synovial membrane only. Intended to modify joint-capsule mechanics; not indicated for tendon or ligament tissue.
Labeled / indicated uses Where each is used	Studied in a mixed real-world population of joint, tendon, and ligament conditions in performance horses across 8 U.S. equine specialty practices.	Registered internationally (AU/NZ 2020) for intra-articular treatment of non-infectious lameness in horses. Joint use only. Not FDA-approved as a drug in the U.S. (Vet Practice Mag; Arthramid label)
Tissue scope in evidence Joints vs. tendons vs. ligaments	Joints, tendons, and ligaments reported in the same pragmatic controlled trial (195 conditions).	Joints only in published evidence. No tendon or ligament indication. (Arthramid clinical studies page)
Sourcing / composition Where the material comes from	Sourced exclusively through partnership with Coteau Grove Farms — closed herd, traceable from foaling to finished product.	Chemically synthesized polymer (polyacrylamide) cross-linked by patented IL-X chemistry; not derived from animal tissue. (Arthramid technical materials)
Peer-reviewed evidence Anchor publications	Characterization & 42-day biocompatibility (BMC Medicine 2025); EV/exosome characterization (ACVSMR 2026 poster); Bertone et al. 2026 pragmatic controlled multicenter clinical trial (Frontiers Vet Sci).	Multiple published studies. Anchor: de Clifford et al. 2021 JEVS double-blinded positive-control study vs. triamcinolone & sodium hyaluronate in racing Thoroughbred middle carpal joints (n≈33). Additional pilot and review publications. (de Clifford 2021; Tnibar 2022)
Study design context How the evidence was generated	Pragmatic multicenter controlled trial: 138 horses / 195 conditions / 8 practices; natural-disease enrollment; open-label; active-comparator vs. clinician-selected standard of care; blinded statisticians & imaging readers.	Single-site double-blinded RCT (de Clifford 2021) plus preclinical osteochondral-fragment model (Contino/Goodrich) and mechanism/histology studies. Higher bias control in the anchor RCT; smaller scale; single joint / single discipline.
Clinical outcome signals How improvement is characterized	Faster and greater improvement in lameness, pain, and swelling vs. SOC; arm-level return-to-performance 52.9% vs. 29.3% at 26 wk (OR 2.72; $p<0.006$). Owner-reported body condition, coat, overall health, and QoL also favored eCTM ($p<0.03$).	Anchor RCT: 83.3% lameness resolution at 6 wk in the 2.5% iPAAG group vs. triamcinolone & HA in racing Thoroughbred middle carpal joints. Broader case-series data cite ~79–82% lame-free at 6–24 mo. (de Clifford 2021; Tnibar 2022)
Onset & durability Timing profile	Separation from SOC by week 2 (pain, swelling) and week 4 (lameness); sustained through 12 wk ($p<0.001$); performance signal durable at 26 weeks.	Onset typically 2–4 weeks (integration window); durability reported at 6–12 months and up to 24 months in some series. Top-up dose at 4–6 wk noted in label if response is partial.
Dosing & workflow Rep-relevant practical facts	Off-the-shelf, ready-to-use, single-dose in most cases. No blood draw, no processing time. Immune-privileged allogeneic tissue.	Off-the-shelf, single 1 mL intra-articular injection per joint; 48-hour rest and ~2-week reduced-impact workload recommended by the label; top-up at 4–6 wk if partial response. (Arthramid vet page)

Category	equicenta® CTM	Arthramid® Vet
Safety profile Comparative safety	Peer-reviewed 42-day biocompatibility (BMC Medicine 2025). In Bertone 2026, eCTM had fewer injection-related adverse events than the clinician-selected SOC arm (standardized public figure: 4.8× fewer injection reactions).	Multi-species safety record cited across published studies; foreign-body / macrophage-driven tissue response is expected and part of the MOA. Notably, in Bertone 2026 the polyacrylamide gel used in the SOC arm was the single largest contributor to injection reactions (including one euthanasia). This finding is specific to the trial's SOC arm and product/site not identified; do not extrapolate to any specific commercial product.
Regulatory status (U.S.) How each is positioned	Working with FDA CVM to open the NADA; product currently manufactured under cGTP with transition to cGMP by end of Q3.	Commercially distributed in the U.S.; not FDA-approved as a drug. Registered internationally in AU/NZ (2020) and used in additional markets under local regulatory pathways.
Reversibility What happens if it doesn't work	Biodegradable — matrix is metabolized as it delivers bioactives; nothing left behind long-term.	Designed to persist in the synovial membrane; product literature positions long-term integration as a feature, but a permanent synthetic component is worth discussing with owners considering future surgical or biologic interventions.

When to lead with equicenta® CTM

- Case involves tendon or ligament — Arthramid is not indicated outside the joint.
- Case involves multiple structures at once (joint + soft tissue) — single-modality biologic handles both.
- Owner or vet prefers a biologic that biodegrades and leaves nothing permanent behind, particularly in younger sport horses with a long career ahead.
- Poor autologous quality suspected (older horse, metabolic disease, prior heavy treatment) — an off-the-shelf standardized allograft removes patient-side variability.
- Case may need future arthroscopy, surgical repair, or additional biologic layering — a synthetic permanent scaffold could complicate that planning conversation.

When Arthramid® Vet may be the better fit

- Isolated, mechanical-driven joint pain in a joint where capsular stiffness is the dominant driver and soft tissue is quiet.
- Chronic OA in an older horse where the goal is comfort and mechanical support rather than tissue-level repair.
- Client explicitly prefers a non-biologic, non-drug device and understands the material integrates and persists in the synovium.
- Cases previously described in the anchor RCT population — middle carpal joint OA in racing Thoroughbreds and comparable presentations.
- Rehab plan can accommodate the 2–4 week onset window and the 48-hour rest + ~2-week reduced-impact workload the label recommends.

30-second positioning: "equicenta CTM and Arthramid Vet are often considered for the same lameness cases but they're fundamentally different tools. Arthramid is a synthetic hydrogel that integrates into and persists in the synovial membrane to change joint mechanics. equicenta CTM is a biologic umbilical-tissue matrix that supports tissue repair, biodegrades, and has published clinical evidence across joints, tendons, and ligaments in a 138-horse, 8-practice controlled trial. If the case is joint-only mechanical OA in an older horse, Arthramid can be a reasonable option. If tendon or ligament is involved, if the horse is a younger sport horse, or if you want a biologic with a broader tissue signal and nothing left behind, lead with equicenta CTM."

APPROVED WORDING RULES (Arthramid conversations)

SAY	DON'T SAY
"Arthramid is a synthetic hydrogel device; equicenta CTM is a biologic tissue matrix. Different tools for different cases."	"Arthramid is unsafe" or "Arthramid doesn't work" — Arthramid has published efficacy in its indicated joint use.
"Arthramid is indicated for joints only; equicenta CTM has published evidence across joints, tendons, and ligaments."	"equicenta CTM replaces Arthramid" — position as an alternative with a different mechanism, not a replacement.
"equicenta CTM biodegrades; Arthramid is designed to integrate and persist in the synovium."	"Arthramid stays in the joint forever" — say "is designed to persist in the synovial membrane" per its label.
"In our multicenter controlled trial, eCTM outperformed clinician-selected standard of care on lameness, pain, swelling, and return to performance."	"eCTM outperformed Arthramid head-to-head" — the Bertone trial did not run a direct head-to-head vs. Arthramid; polyacrylamide gel appeared inside the SOC arm without brand identification.
"Arthramid product/brand identity in the Bertone SOC arm was not reported; the injection-reaction observation is trial-specific."	Naming any specific commercial polyacrylamide product as the source of the Bertone injection-reaction signal.

Sources. Arthramid product characterization, MOA, and label positioning: arthramid.com/equine-vets, Arthramid clinical studies index, Vet Practice Mag 2024, and Tnibar 2022 review. Anchor RCT: de Clifford et al. 2021, JEVS. Independent PICO review: Equine osteoarthritis: is intra-articular polyacrylamide gel better than intra-articular corticosteroid?. equicenta CTM clinical evidence: Bertone AL, et al. 2026, Frontiers in Veterinary Science, DOI 10.3389/fvets.2026.1833933. Characterization & 42-day biocompatibility: BMC Medicine 2025. EV/exosome characterization: ACVSMR 2026 poster.